Task 5 : Capture and Analyze Network Traffic Using Wireshark.

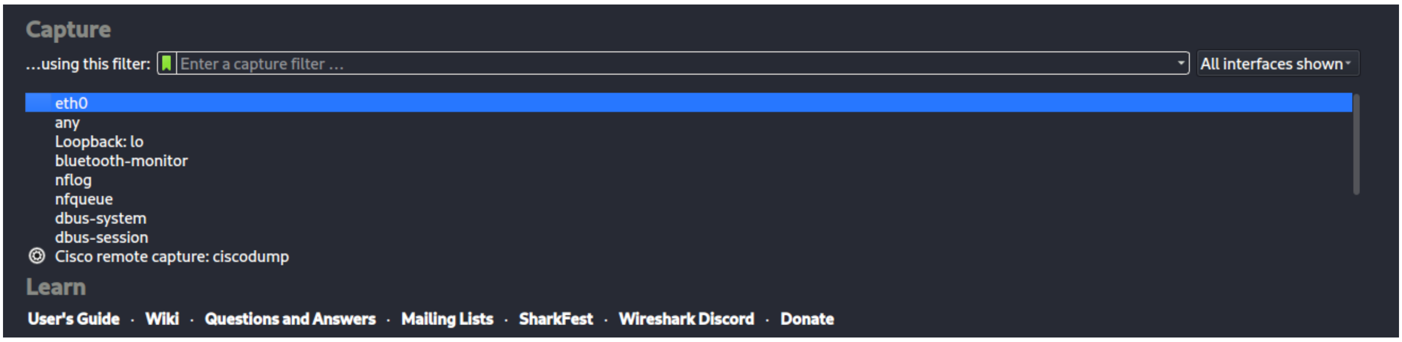
Objective: Capture live network packets and identify basic protocols and traffic types.

Tools: Wireshark (free).

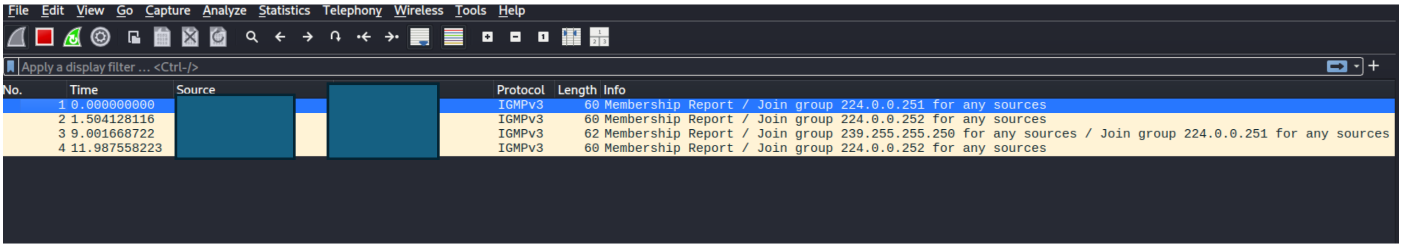
Deliverables: A packet capture (.pcap) file and a short report of protocols identified.

Steps:-

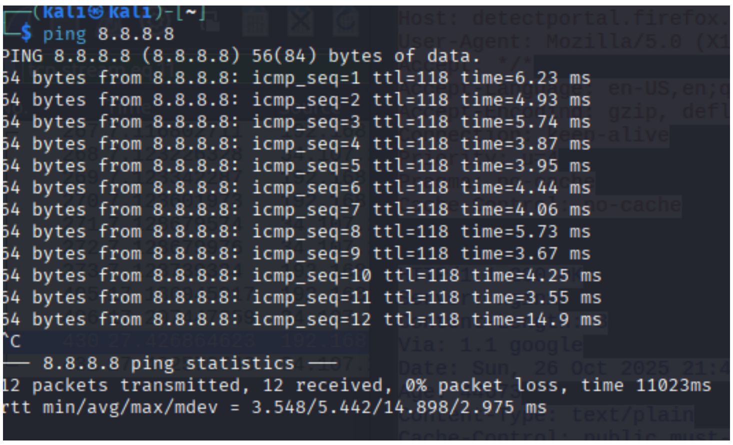
1. Start capturing on your active network interface



2. .Browse a website or ping a server to generate traffic



3. ping google.com



4. Filter captured packets by protocol (TCP)

```
GET /success.txt?ipv4 HTTP/1.1
Host: detectportal.firefox.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Priority: u=4
Pragma: no-cache
Cache-Control: no-cache


HTTP/1.1 200 OK
Server: nginx
Content-Length: 8
Via: 1.1 google
Date: Sun, 26 Oct 2025 21:40:10 GMT
Age: 44673
Content-Type: text/plain
Cache-Control: public,must-revalidate,max-age=0,s-maxage=3600

success
```

5. Identify at least 3 different protocols in the capture.

Protocol
IGMPv3
NTP
NTP
ICMP
ICMP
ARP
ARP
ICMP
ICMP
IGMPv3
ICMP
ICMP
ICMP
ICMP
ICMP
ICMP
ICMP
ICMP

6. Summary:-

TCP (Transmission Control Protocol):

- Used for establishing reliable connections between client and server.
- Observed during communication with web servers for data transfer.

HTTP (Hypertext Transfer Protocol):

- Application layer protocol for web traffic.
- Detected when accessing websites — showing GET/POST requests and responses.

ICMP (Internet Control Message Protocol):

- Used for network diagnostics (e.g., ping command).
- Observed ICMP Echo Request and Echo Reply packets for connectivity testing.